

A Project Report
On
**Intelligent Diabetes Clinical Decision
Support System using Causal Inference and
RAG**

Submitted in partial fulfilment of the requirement of
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COMPUTER ENGINEERING

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APPROVAL SHEET

This is to certify that the project entitled

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Declaration

We declare that this written submission for the B.Tech Major Project entitled “Intelligent Diabetes Clinical Decision Support System using Causal Inference and RAG” represents our ideas in our own words, and where others’ ideas or words have been included, we have adequately cited and referenced the original sources. We also declare that we have adhered to all principles of academic honesty and integrity and have not misrepresented, fabricated or falsified any idea, data, fact or source in our submission. We understand that any violation of the above will be cause for disciplinary action by the Institute and can also evoke penal action from the sources which have not been properly cited or from whom proper permission has not been taken when needed.

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Abstract

India carries one of the largest diabetes burdens in the world, and when an adult with type 2 diabetes is not controlled on metformin, the clinician must choose a second-line drug with little patient-specific evidence. Existing digital tools either restate guideline rules, predict risk from cross-sectional data such as the Pima dataset, or answer free-text questions through general-purpose language models that can produce uncited or unsafe content. Predictive models answer “who is at risk” but not “what would happen to this patient under each drug”, which is a causal question distorted by confounding by indication. This work presents DiaCausal, a research prototype that first removes unsafe options with deterministic, cited safety rules, then estimates the six-month HbA1c change under a sodium–glucose co-transporter-2 inhibitor, a dipeptidyl peptidase-4 inhibitor and a sulfonylurea using a cross-fitted multinomial propensity model, augmented inverse probability weighting and a doubly robust learner, each with a 95% interval, and abstains when overlap is insufficient. Because no open Indian dataset records treatment and outcome, the engine is developed and benchmarked on an India-calibrated synthetic cohort with known true effects. Over 20 synthetic cohorts of 5,000 patients, the naive comparison overstated the SGLT2i-versus-DPP-4i effect by about 75%, whereas AIPW reduced the bias to at most 0.005 HbA1c points with 90–95% interval coverage, and the DR-learner’s patient-level intervals covered the truth for 92.7–94.7% of test patients. The engine runs in a live, installable phone website with admin-approved accounts and an early retrieval layer that returns cited passages from licence-cleared sources. This mid-semester report presents the proposed system, requirements, design, timeline, the implementation completed so far, initial results and the evaluation matrix. Intended use: Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

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Chapter 1

Introduction

1.1 Background

Type 2 diabetes is one of India’s largest public-health problems. The ICMR-INDIAB national cross-sectional study of 113,043 adults reported a weighted diabetes prevalence of 11.4% (95% CI 10.2–12.5) and a prediabetes prevalence of 15.3%, corresponding to an estimated 101 million people with diabetes and 136 million with prediabetes in India [2]. Worldwide, the IDF Diabetes Atlas estimates that 589 million adults were living with diabetes in 2024 [3].

Metformin is the usual first-line drug. When it is not enough, the clinician adds a second agent, commonly a sodium–glucose co-transporter-2 inhibitor (SGLT2i), a dipeptidyl peptidase-4 inhibitor (DPP-4i) or a sulfonylurea (SU). These classes differ in glucose-lowering effect, hypoglycaemia risk, effect on weight and cost, and patients do not all respond alike. In the TriMaster crossover trial, participants with an eGFR of 60–90 mL/min/1.73m² had an HbA1c 2.90 mmol/mol (95% CI 1.19–4.61) lower on sitagliptin than on canagliflozin, compared with participants whose eGFR was above 90 [4]. A good decision therefore depends on the individual patient.

1.2 Motivation

Choosing a second-line drug is a counterfactual question: what would this patient’s HbA1c be under each option? Routine records cannot answer it directly, because doctors choose drugs according to the patient’s condition (confounding by indication). Population averages from network meta-analyses [5, 6] do not individualise, and clinical decision support systems can improve care but also carry risks such as alert fatigue, opacity and unsafe output [7]. Large language models can explain, but may invent doses or citations; retrieval-augmented generation (RAG) grounds answers in source documents [8, 9]. DiaCausal combines these ideas in a strict order: safety rules first, then causal estimation with honest uncertainty, then a cited explanation — with the clinician always making the decision.

1.3 Aim and Objective

Aim: To design and build a clinician-in-the-loop decision-support prototype that estimates the individualised six-month HbA1c change for SGLT2i, DPP-4i and sulfonylurea added to metformin, with uncertainty intervals, abstention when evidence is insufficient, and cited explanations.

Objectives:

1. Encode contraindication, kidney-function and interaction rules as structured, cited, unit-tested tables that run before any estimate is shown.
2. Build an India-calibrated synthetic cohort with known true treatment effects, with every parameter’s source recorded.
3. Implement generalised propensity scoring with overlap-based abstention, augmented inverse probability weighting (AIPW) for average effects and a doubly robust learner (DR-learner) for individual effects, all with 95% intervals.
4. Build a licence-compliant RAG layer with hybrid retrieval and section-level citations that refuses to answer out of scope.

5. Evaluate the system with the evaluation matrix of Chapter 5, including a clinician vignette study.

1.4 Report Outline

Chapter 2 reviews the literature and existing systems. Chapter 3 states the problem, the scope and the proposed system. Chapter 4 presents requirement engineering, the hardware and software requirements, and the system design. Chapter 5 reports the implementation completed so far, the initial results, the demo and the evaluation matrix. Chapter 6 concludes and describes the planned work. Appendix A gives the timeline chart and Appendix B the publication plan.

Chapter 2

Study Of the System

2.1 Literature Survey

Treatment selection in type 2 diabetes. Shields et al. ran TriMaster, a randomised, double-blind, three-way crossover trial of sitagliptin, canagliflozin and pioglitazone in 525 participants, and showed that simple features such as BMI and eGFR predict which drug lowers HbA1c more [4]. Dennis et al. developed an algorithm that selects between SGLT2 and DPP-4 inhibitors from routine UK primary-care data [10]. Güdemann et al. validated that algorithm across UK ethnic groups and found that the average predicted HbA1c benefit of SGLT2i over DPP-4i was 2.1 mmol/mol (95% CI 1.6–2.6) in South Asian patients compared with 3.7 mmol/mol (3.5–3.9) in White patients, recommending recalibration before use in new populations [11] — direct evidence that calibration for Indian patients matters.

Causal-inference methods. The potential-outcomes framework [12], structural causal models [13] and target-trial thinking [14, 15] define what is being estimated. Meta-learners [16], causal forests [17], the DR-learner [18] and double machine learning [19] estimate effects that vary between patients. Semi-synthetic benchmarks introduced the PEHE error measure [20, 21], and the open-source libraries DoWhy [22] and EconML [23] implement many of these estimators.

RAG in medicine. Lewis et al. introduced retrieval-augmented generation [8]. Almanac improved factuality and safety over a standalone language model on clinical questions [9], and the MedRAG toolkit with the MIRAGE benchmark compared many corpus, retriever and model combinations [24]. RAGAS provides reference-free metrics such as faithfulness for evaluating such systems [25].

Causal methods applied in health. Quaranta and Pietrantuono applied causal discovery and DoWhy counterfactual analysis to the Pima dataset of 768 female patients [26]. Hasan et al. used an ensemble of causal trees to formulate oxygen-therapy policy for ventilated patients in MIMIC-III [27]. Song et al. used causal discovery with heart rate as a confounder in a hierarchical regression model for cuffless blood-pressure estimation [28].

Table 2.1: Comparison of the reviewed literature

Paper	Data	Approach	Gap for DiaCausal
Shields et al. 2023 [4]	TriMaster RCT, UK	Stratified three-way crossover trial	Two simple strata; UK population
Dennis et al. 2022 [10]	UK primary care	Treatment-selection model, SGLT2i vs DPP-4i	Two drugs only; no sulfonylurea; not Indian
Güdemann et al. 2026 [11]	UK primary care	Validation across ethnic groups	Smaller South Asian benefit; needs recalibration
Quaranta and Pietrantuono 2025 [26]	Pima (768)	DirectLiNGAM and DoWhy	Not Indian; no treatment variable
Hasan et al. 2025 [27]	MIMIC-III	Heterogeneous effects with causal trees	Single ICU dataset; not clinically validated
Song et al. 2024 [28]	VitalDB + 17 subjects	Causal discovery, hierarchical regression	Not diabetes; linear assumptions
Zakka et al. 2024 [9]	314 clinical questions	RAG over curated sources	No patient-specific causal estimates
Xiong et al. 2024 [24]	MIRAGE benchmark	Benchmarking medical RAG	Question answering, not decision support

2.2 Existing systems

Four kinds of systems address parts of this problem today.

Rule-based guideline tools and calculators present recommendations as flowcharts or scores. They are transparent but give the same answer to every patient in a category and do not estimate how much a drug would help this patient.

Predictive machine-learning systems, often built on the Pima dataset, predict the risk of diabetes. They answer “who is at risk”, not “what would happen under each drug”, and Pima comes from the Akimel O’odham community in Arizona, USA, so it is not Indian data and records no treatment.

General-purpose language-model chatbots answer free-text questions fluently but cannot guarantee sources, may invent facts or doses, and do not produce patient-specific causal estimates.

Research treatment-selection models such as the SGLT2i/DPP-4i algorithm [10] and the TriMaster strata [4] individualise the choice, but compare two drugs, are built on UK data, and do not include safety rules, abstention or cited explanations.

Table 2.2: Feature comparison of existing systems and DiaCausal (Y = yes, P = partly, N = no)

Feature	Rules	Pima ML	Chatbot	Selection models	DiaCausal
Per-patient effect of each drug	N	N	N	P	Y
Uncertainty intervals	N	P	N	Y	Y
Refuses when data is thin	N	N	N	N	Y
Safety rules before any estimate	Y	N	N	N	Y
Cited guideline explanation	P	N	P	N	Y
Indian calibration (cut-offs, INR cost)	P	N	N	N	Y
Audit log of every request	P	N	N	N	Y
Clinician makes the decision	Y	P	N	Y	Y

Chapter 3

Proposed System

3.1 Problem Statement

For an adult with type 2 diabetes who is already on metformin and inadequately controlled, estimate the six-month HbA1c change under an SGLT2 inhibitor, a DPP-4 inhibitor and a sulfonylurea — each with an uncertainty interval — after deterministic safety rules have removed the options that are unsafe for that patient, and explain the result with citations to approved guidelines, leaving the decision to the clinician.

3.2 Scope

In scope. Adults with type 2 diabetes on metformin; three second-line options; primary outcome the six-month change in HbA1c (%); secondary outcomes hypoglycaemia risk, weight change (kg) and monthly cost (INR); users are clinicians, evaluators and an administrator.

Out of scope (future work). Insulin and GLP-1 receptor agonists; diagnosis; dosing advice of any kind; use on real patients without ethics approval; causal discovery as the main method (DirectLiNGAM or PC algorithms may only be used to sanity-check the expert-drawn graph).

Scope refinement. The first proposal targeted hospital readmission using the Diabetes 130-US Hospitals dataset, and the second targeted early diagnosis on the Pima, NHANES and MIMIC-III datasets. A dataset review in September 2026 showed that the Diabetes 130-US data predates SGLT2 inhibitors and that the Pima dataset (from the Akimel O’odham community in Arizona, USA; not Indian; no treatment variable; zeros used for missing values) cannot answer a treatment question. The team therefore froze the present scope.

3.3 Proposed System

3.3.1 Why causal inference: a worked example

The following illustrative synthetic numbers show why a simple comparison misleads (Table 3.1).

Table 3.1: Confounding by indication (illustrative synthetic numbers; HbA1c change in %)

Patients	Given SGLT2i	Given sulfonylurea	Difference
High starting HbA1c	300 patients, -1.2	100 patients, -1.0	-0.2
Lower starting HbA1c	100 patients, -0.6	300 patients, -0.5	-0.1
All patients (naive)	-1.05	-0.625	-0.425

Comparing like with like and weighting the two groups equally gives $0.5(-0.2) + 0.5(-0.1) = -0.15$. The naive difference of -0.425 is about three times too large only because patients with higher HbA1c received SGLT2i more often.

3.3.2 Formulation

Let X be a patient’s baseline characteristics, $T \in \{\text{SGLT2i}, \text{DPP-4i}, \text{SU}\}$ the drug added, and $Y(t)$ the potential six-month HbA1c change under drug t . The quantity of interest for two drugs j and k is the conditional average treatment effect

$$\tau_{jk}(x) = E[Y(j) - Y(k) \mid X = x]. \quad (3.1)$$

It can be estimated from observational data under three assumptions: consistency; conditional exchangeability given the adjustment set from the causal graph; and positivity, $e_t(x) = P(T = t \mid X = x) > \varepsilon$. The average outcome under drug t is estimated by augmented inverse probability weighting (AIPW) with five-fold cross-fitting:

$$\hat{\mu}_t = \frac{1}{n} \sum_{i=1}^n \left[\hat{m}_t(X_i) + \frac{\mathbf{1}\{T_i = t\} (Y_i - \hat{m}_t(X_i))}{\hat{e}_t(X_i)} \right], \quad (3.2)$$

where $\hat{m}_t$ is a gradient-boosted outcome model and $\hat{e}_t$ the multinomial logistic propensity model, both fitted out of fold. The 95% interval of an average effect uses the standard error of the per-patient scores (their influence function). The DR-learner regresses the difference of these pseudo-outcomes, $\hat{\varphi}_j - \hat{\varphi}_k$ with $\hat{\varphi}_t = \hat{m}_t(X) + \mathbf{1}\{T = t\}(Y - \hat{m}_t(X))/\hat{e}_t(X)$, on X with a linear second stage [18]; each patient’s 95% interval comes from its heteroscedasticity-robust (HC3) covariance. If an allowed drug has $\hat{e}_t(x) < 0.05$ for this patient, the system shows “insufficient evidence” for it instead of a number. Sensitivity to unmeasured confounding is summarised by the E-value $E = RR + \sqrt{RR(RR - 1)}$.

3.3.3 Variables and causal graph

The variables are chosen with clinical knowledge, not discovered from the data (Figure 3.1). The engine adjusts for age, sex, diabetes duration, baseline HbA1c, eGFR, BMI, established cardiovascular disease, heart failure, past hypoglycaemia, past ketoacidosis, past pancreatitis and affordability (cost concern). Baseline HbA1c, eGFR and sex modify the effect. Excluded because they occur after treatment starts: weight change (a mediator), adherence, three-month laboratory values, drug switching (a collider) and hospitalisation during follow-up. Prescriber preference is also excluded because it predicts the drug but not the outcome, and adjusting for it amplifies bias. BMI categories use Asian-Indian cut-offs (overweight ≥ 23 and obesity ≥ 25 kg/m²) [29].

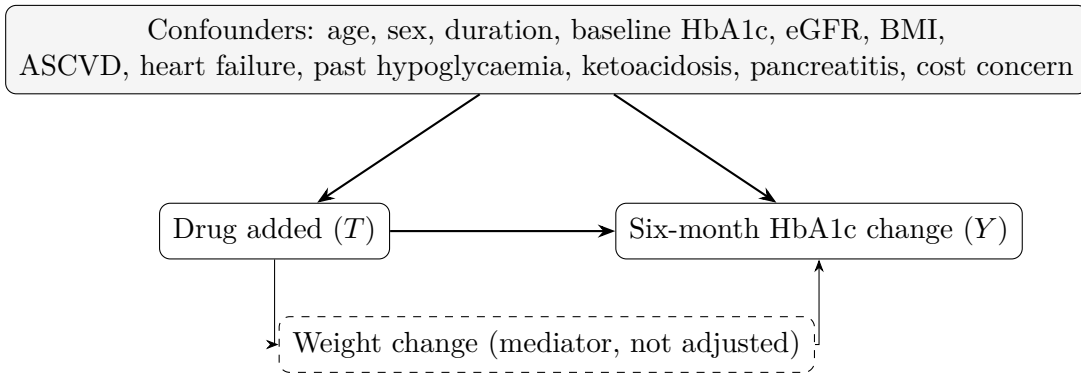


Figure 3.1: Expert causal graph (confounders grouped for clarity)

3.3.4 Pipeline and safety rules

The request pipeline is: patient input $\rightarrow$ safety rules $\rightarrow$ propensity and overlap check $\rightarrow$ AIPW and DR-learner estimates with intervals $\rightarrow$ cost lookup $\rightarrow$ cited explanation (RAG) $\rightarrow$ clinician decision $\rightarrow$ audit log. The system follows eight rules that are never broken:

1. Decision support only; the clinician always decides.
2. Safety rules run before any estimate is shown.
3. Doses, thresholds and contraindications come only from structured, cited tables — never from a language model. Doses are never shown.
4. The system abstains rather than guesses.
5. Every estimate carries an interval.
6. Every clinical claim cites a specific source, version and section.
7. Every request is logged and versioned; patient values are never written to the log.
8. Intended use: Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

3.3.5 Data strategy

No openly downloadable, patient-level Indian dataset records which second-line drug a patient received and what happened afterwards. The system therefore uses three steps. (1) Calibrate with published Indian figures from ICMR-INDIAB [2] and the Longitudinal Ageing Study in India [30]. (2) Build an India-calibrated synthetic cohort with known true effects, recording each parameter’s source and status (CITED, ASSUMED-DIRECTIONAL or TEAM-SET); drug-effect directions follow network meta-analyses [5, 6] and treatment-selection studies [10, 4, 11]. (3) Validate against the known truth now, and later against real data and a clinician review of realistic cases. Real Indian clinic data will be used only with ethics committee approval.

Chapter 4

Design Of the System

4.1 Requirement Engineering

4.1.1 Requirement Elicitation

Requirements were gathered from meetings with the project guide, discussions with a collaborating clinician, the drug labels and guidelines used for the safety rules, the team’s dataset review and variable-selection notes (September 2026), and the literature in Chapter 2. They were written as numbered, testable statements following the ISO/IEC/IEEE 29148 style.

4.1.2 Software lifecycle model

The project follows an incremental, iterative model with one-week sprints and a review gate at the end of each week. Each increment ends with working, tested software: the causal engine, the phone website and a first RAG layer with its evaluation for the mid-semester review (30 September), search improvements and integration (9 and 16 October), clinician evaluation (23 October) and the final version (30 October). A waterfall model would leave all integration risk to the end; the automated safety tests (202 at present, plus 647 for the chat application) act as a gate on every increment, and continuous integration runs them on Linux, macOS and Windows for every change.

4.1.3 Requirement Analysis

Table 4.1: Functional requirements and their status on 26 September 2026

ID	Requirement	Status
FR1	Capture age, sex, diabetes duration, BMI, baseline HbA1c (%), eGFR (mL/min/1.73m ²), cardiovascular disease, heart failure, past hypoglycaemia, ketoacidosis, pancreatitis, type 1 diabetes and cost concern.	Done
FR2	Validate inputs and units; reject out-of-range values with a clear message.	Done
FR3	Apply the cited safety rules and show every exclusion or caution with its reason and source.	Done (10 rules)
FR4	Compute generalised propensity scores and an overlap check for each remaining option.	Done
FR5	Show “insufficient evidence”, with the reason, when overlap is too low.	Done
FR6	Estimate each allowed option’s six-month HbA1c change with a 95% interval.	Done
FR7	Show hypoglycaemia risk and weight change with intervals.	Done
FR8	Look up the monthly cost (INR) from a dated, cited price table.	Done; prices await confirmation
FR9	Show a cited explanation from licence-cleared sources only.	Done (first version)
FR10	Log every request: model and rule versions, statuses and rule IDs (never patient values).	Done

Table 4.1: Functional requirements and their status on 26 September 2026

ID	Requirement	Status
FR11	Export a consultation summary.	Done (print / save as PDF)
FR12	Sign-up, authenticator-app sign-in and admin approval for evaluators.	Done

Table 4.2: Non-functional requirements

Quality	Requirement
Clinical safety	All eight safety rules enforced by automated tests
Correctness	Estimator tests against known true effects on synthetic data
Performance	Under 3 seconds per request on a laptop (the website answers on the phone itself)
Availability	Demo-grade; installable website that works offline, plus a recorded demo
Security	HTTPS, sign-in with an authenticator app, admin approval, strict content-security policy
Privacy	No identifiable patient data; patient details never leave the device; principles of India’s DPDP Act, 2023
Auditability	Append-only log of every request
Usability	System Usability Scale score of at least 68 [31]
Accessibility	WCAG 2.1 AA for the web interface
Maintainability	Readable, typed Python; every module tested
Licence compliance	Every knowledge source recorded with its licence category

4.1.3.1 UML diagrams/ DFDs based on the project

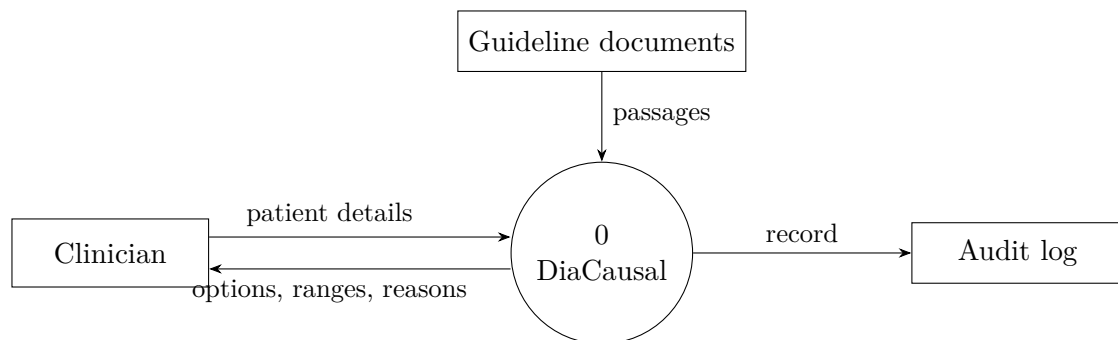


Figure 4.1: Data flow diagram, level 0

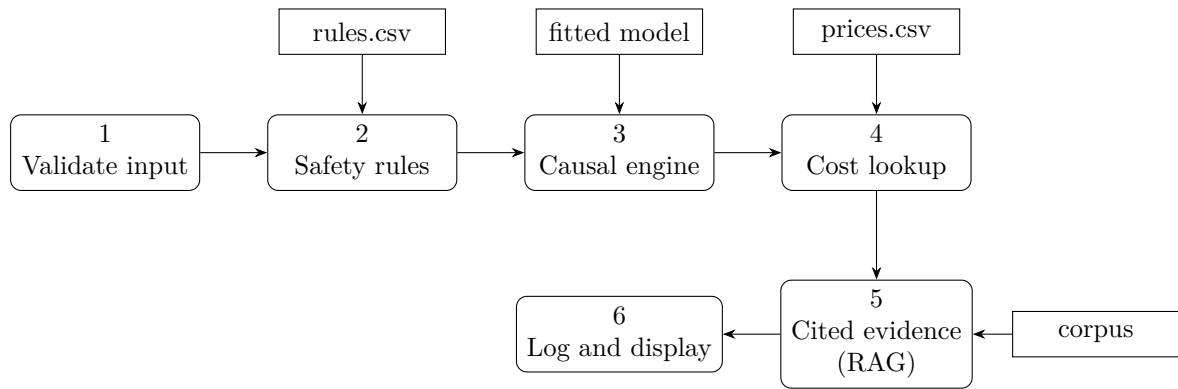


Figure 4.2: Data flow diagram, level 1

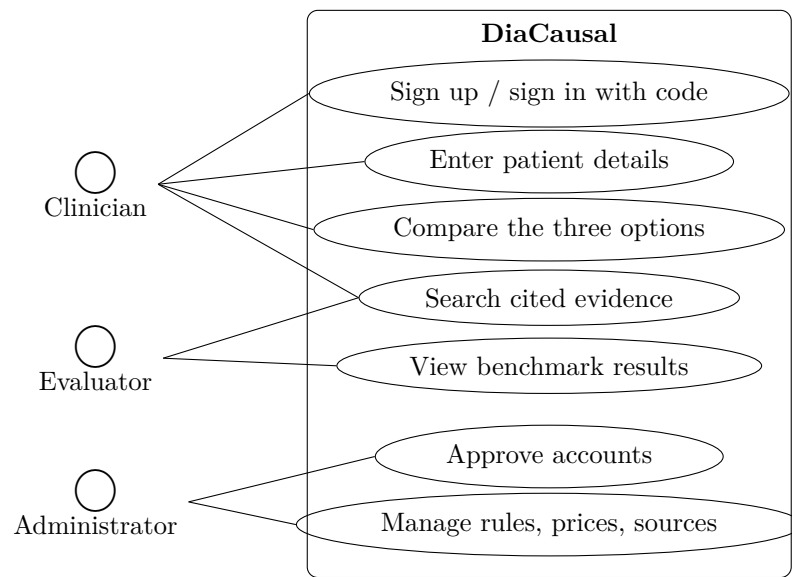


Figure 4.3: Use-case diagram

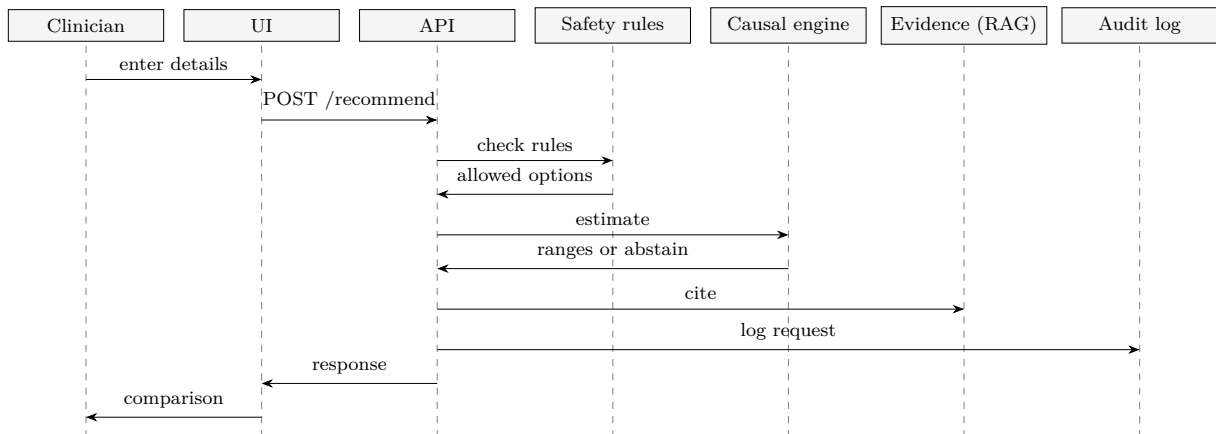


Figure 4.4: Sequence diagram for one consultation

Data design. The main records are:

- REQUEST: id, time, model version, parameter version, rule version (no patient values);

- ARM RESULT: request id, drug, status, estimate, lower and upper limit, propensity;
- GUARDRAIL RULE: rule id, drug, field, operator, value, action, message, source, section, status;
- DRUG COST: drug, INR per month, as-of date, source;
- SOURCE REGISTRY (RAG): source id, title, version, licence category, checked by;
- PROFILE (accounts): name, email, role, approved, admin, intended-use acknowledgement date.

API design. POST `/api/v1/recommend` accepts the patient details and returns, for each drug, the estimate with its 95% interval or “insufficient evidence”, any exclusions and cautions with sources, fair pairwise comparisons, the assumptions, the version strings, a request id and the intended-use statement. Supporting endpoints are GET `/api/v1/rules` and GET `/health`.

4.1.3.2 Cost Analysis

All software is open source, so the licence cost is zero. Development uses the team’s own laptops. The website is hosted on Netlify’s free tier and the accounts on Supabase’s free tier (Mumbai region), so the running cost is zero. The RAG explanation writer uses a template (free, offline), Google’s Gemini free tier through a Supabase Edge Function, or a local Ollama model; no paid service is required. The main cost is effort: about four members $\times$ 12 hours per week $\times$ 14 weeks, roughly 670 person-hours (team estimate).

4.1.3.3 Hardware and software requirement

Table 4.3: Hardware and software requirements

Tier	Hardware	Software
Development	Laptop with at least 8 GB RAM (e.g. MacBook Air, Apple silicon)	Python 3.12, NumPy, pandas, scikit-learn, Matplotlib, pytest, Node.js 24, Git and GitHub
Causal engine and API	The same laptop	FastAPI, Uvicorn, Pydantic, Streamlit (demo)
Website	Any phone or laptop with a modern browser	HTML, CSS, JavaScript (installable, offline-capable); Netlify hosting; Supabase accounts
Chat application	Laptop, or a 2-CPU/4 GB server	FastAPI, React, TypeScript, SQLite, faster-whisper, Docker
RAG	Same laptop	PDF parser, BM25 and TF-IDF search, rank fusion; template, online or local language model
Documentation	Any laptop	LaTeX (Overleaf), presentation software

4.2 System architecture

4.2.1 UI/UX diagram

The interface is a single consultation screen (Figure 4.5). Colours carry meaning and never suggest a prescription: red means excluded, amber means caution, grey means insufficient

evidence, and green is never used for “recommended”. The intended-use statement appears at the top of every screen. The layout is phone-first (iPhone 15 size) and widens to three columns on a laptop.

Category: Obese (25 or above, Asian-Indian cut-off)

History

- ☐ Established heart disease (ASCVD)
- ☐ Past hypoglycaemia
- ☒ Past pancreatitis
- ☐ Cost is a major concern
- ☐ Heart failure
- ☐ Past diabetic ketoacidosis
- ☐ Type 1 diabetes

Compare the three options

BMI category: Obese (25 or above, Asian-Indian cut-off) · Request ID 85769827a5b8 · computed on this device

The three options (safety rules ran first)

SGLT2 inhibitor	DPP-4 inhibitor	Sulfonylurea
e.g. dapagliflozin	e.g. sitagliptin	e.g. glimepiride
EXCLUDED	ESTIMATE · CAUTION	ESTIMATE · CAUTION
Not estimated: removed by a safety rule before the causal engine ran.	-0.68 (95% range -1.09 to -0.27)	-0.82 (95% range -1.21 to -0.42)
R01 · Excluded (egfr < 45; rule verified): Not recommended to improve glycaemic control when eGFR < 45 mL/min/1.73 m2. KDIGO 2022 Rec 1.3.1 supports SGLT2i at eGFR ≥ 20 for kidney/heart protection - a different purpose; flag for the clinician.	percentage points of HbA1c at 6 months · propensity 0.54	percentage points of HbA1c at 6 months · propensity 0.23
Source: FARXIGA (dapagliflozin) US prescribing information (DailyMed); KDIGO 2022 Diabetes in CKD — PI Sec 1 Limitations of Use, 2.2; KDIGO Rec 1.3.1 Cost: price unavailable	R04 · Caution (egfr < 45; rule verified): Renal dose adjustment required per label when eGFR < 45 mL/min/1.73 m2. This system never displays doses - consult the label. Source: JANUVIA (sitagliptin) US prescribing information (DailyMed) — Sec 2.2	R09 · Caution (egfr < 60; rule unverified): Kidney impairment increases hypoglycaemia risk; label advises a conservative start. eGFR cut-off 60 is team-set. Source: Glimepiride US prescribing information (DailyMed) — Sec 2.1; 8.6 Cost: price unavailable

Figure 4.5: Consultation screen: patient details, then the three options with exclusions, cautions and sources

4.2.2 Block Diagram

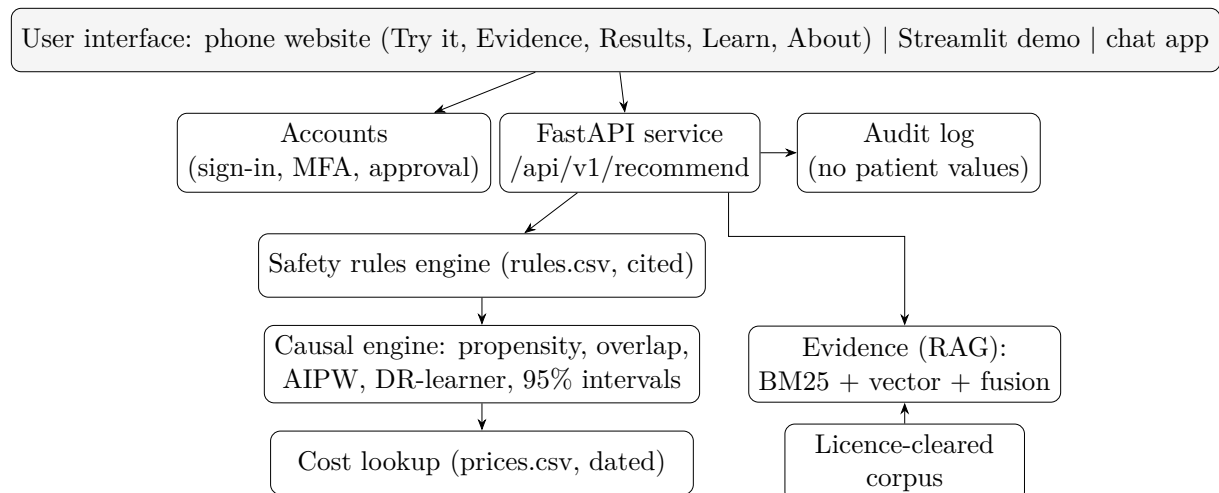


Figure 4.6: Block diagram of DiaCausal

The website runs the same fitted model and the same evidence search inside the browser (exported from Python as `model.json` and `evidence.json`); automated tests check that the browser and Python give identical answers, so patient details never need to leave the device.

Chapter 5

Result and Discussion

5.1 Implementation status

Table 5.1 lists every module and its status on 28 September 2026. All code is in the public repository github.com/Rhian-Roy/DiaCausal-CDSS; 201 automated tests (149 engine, 33 RAG, 19 website) and the chat application’s 630 tests (392 backend, 238 frontend) pass on every change, on Linux, macOS and Windows.

Table 5.1: Implementation status (28 September 2026)

Module	Status	Evidence	Owner
Synthetic cohort generator	Done	<code>params.yaml</code> : every number has a source and a status; tests	Rhian
Safety rules (<code>rules.csv</code>)	Done	10 cited rules (7 verified against labels, 3 team-set cut-offs flagged for the clinician); tests	Rhian
Propensity model and overlap check	Done	Cross-fitted multinomial model; overlap figure; tests	Rhian
Naive, IPW, matching and AIPW	Done	Results table; 95% intervals; tests	Rhian
DR-learner (patient-level effects)	Done	Recovery and calibration figures; S- and T-learner baselines	Rhian
Benchmark, metrics, refutation, E-values	Done	<code>results/</code> : 20 cohorts $\times$ 5,000 patients; 9/9 refutation checks pass	Rhian
Demo (Streamlit) and FastAPI endpoint	Done	POST <code>/api/v1/recommend</code> ; screenshots	Rhian
Phone website (installable, offline)	Done	https://diacausal.netlify.app ; browser and Python give identical answers (155 patients)	Rhian, Grace-ton
Accounts: sign-up, authenticator app, admin approval	Done	Row-level security; tested end to end	Rhian
Chat application (guards, CAPTCHA and MFA sign-in, patient panel, clinical guardrails, voice, Docker)	Done	647 tests; <code>check_all</code> 43 checks	Team
Hypoglycaemia and weight outcomes; too-uncertain abstention; consultation summary	Done	Secondary rows in <code>results/</code> ; print summary on the website	Rhian
RAG retrieval with citations	Done	WHO 2018 + 7 FDA safety communications (78 passages); licence gate; BM25 + TF-IDF + fusion; coverage abstention	Rhian

Table 5.1: Implementation status (28 September 2026)

Module	Status	Evidence	Owner
RAG explanations, gold questions, evaluation	Done (first version)	Quoted / Gemini / Ollama explanations behind a citation checker; 60 gold questions; <code>results/rag_eval_summary.csv</code>	Rhian
Engine and RAG inside the chat application (all six stages)	Done (29 Sep)	9 new pipeline tests; estimates with 95% ranges and cited evidence in the reply	Team
Embedding model, reranker, held-out questions	Planned	Target 3 October	Team
Clinician vignette review and usability survey	Planned	10–23 October	Advik
Research paper	Planned	After results are frozen (16 October)	Pratham

5.2 Implementation

The engine is written in Python and organised as small, separately tested modules. The generator creates an India-calibrated synthetic cohort in which doctors’ prescribing depends on patient characteristics (confounding by indication) and every patient’s outcome under all three drugs is stored, so the true effects are known. The safety rules module reads `rules.csv` and removes or flags options with a cited reason; thresholds are never hard-coded, and a test scans the code to prove it. The propensity module fits a five-fold cross-fitted multinomial logistic regression and flags options whose propensity is below 0.05. The estimators module computes naive, inverse probability weighted (IPW), matching and AIPW average effects with 95% confidence intervals, and a DR-learner with robust intervals for patient-level effects. The metrics module compares every estimate with the known truth. A benchmark script repeats the whole process on 20 simulated cohorts and writes the results table and figures used below. A Streamlit demo, a FastAPI endpoint and the phone website expose the same pipeline, and each reply is checked for the intended-use statement and for dose-like text before it is shown.

5.3 Initial results

All results in this section come from the synthetic cohort, where the true effects are known. They show whether the methods recover the truth; they do not show clinical validity on real patients.

Table 5.2: Synthetic benchmark results: 20 cohorts of 5,000 patients, with 2,000 fresh test patients each

Contrast	Method	Truth	Bias	RMSE	95% CI coverage
SGLT2i vs DPP-4i	naive	-0.211	-0.159	0.160	0.00
SGLT2i vs DPP-4i	IPW	-0.211	0.001	0.022	1.00
SGLT2i vs DPP-4i	matching	-0.211	-0.002	0.026	0.95
SGLT2i vs DPP-4i	AIPW	-0.211	0.002	0.023	0.95
Sulfonylurea vs DPP-4i	naive	-0.256	-0.106	0.108	0.00
Sulfonylurea vs DPP-4i	IPW	-0.256	-0.004	0.025	1.00
Sulfonylurea vs DPP-4i	matching	-0.256	-0.008	0.028	1.00
Sulfonylurea vs DPP-4i	AIPW	-0.256	-0.003	0.027	0.95
SGLT2i vs Sulfonylurea	naive	0.045	-0.052	0.056	0.50
SGLT2i vs Sulfonylurea	IPW	0.045	0.005	0.030	0.95
SGLT2i vs Sulfonylurea	matching	0.045	0.006	0.026	0.95
SGLT2i vs Sulfonylurea	AIPW	0.045	0.005	0.030	0.90

Learner	PEHE SGLT2i vs DPP-4i	PEHE Sulfonylurea vs DPP-4i	PEHE SGLT2i vs Sulfonylurea	Policy regret
DR-learner	0.105	0.117	0.108	0.011
T-learner	0.189	0.192	0.184	0.029
S-learner	0.103	0.104	0.110	0.012
naive	0.214	0.197	0.169	0.062

The true effects are -0.211 (SGLT2i minus DPP-4i), -0.256 (sulfonylurea minus DPP-4i) and $+0.045$ (SGLT2i minus sulfonylurea) HbA1c percentage points. The naive comparison overstates the SGLT2i-versus-DPP-4i benefit by about 75% (-0.370 against -0.211) and its intervals never contain the truth, because sicker patients received SGLT2i more often. IPW, matching and AIPW remove almost all of this bias ($|\text{bias}| \leq 0.008$) with 90–100% interval coverage. For individual patients, the DR-learner’s error (PEHE 0.105–0.117) is about half the T-learner’s and similar to the S-learner’s, but unlike either it gives every patient a 95% interval, which contained the true patient effect for 94.4%, 94.7% and 92.7% of test patients. Choosing the option with the lowest DR-learner estimate lost on average 0.011 HbA1c points against perfect knowledge (policy regret), compared with 0.062 for the naive averages. The system answered “insufficient evidence” or “excluded” for 2.5% of patient–option pairs. After weighting, the largest standardised mean difference fell from 0.70 to 0.08, below the 0.1 balance threshold [1]. All nine refutation checks passed (placebo treatment, random common cause and 80% subsets for each contrast), and the E-values were 1.57–2.15 for the estimates.

Average effect: estimate and 95% CI (averaged over repeats) vs the true value

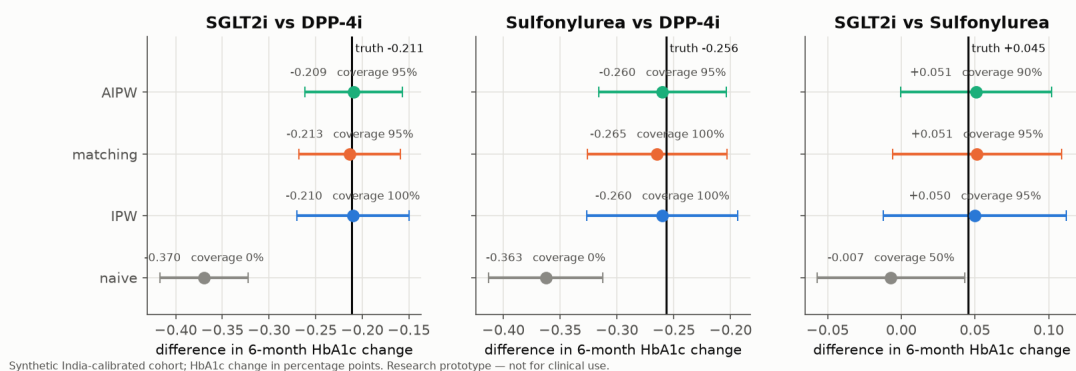


Figure 5.1: Estimated versus true average effect for the naive, IPW, matching and AIPW methods across repeated cohorts

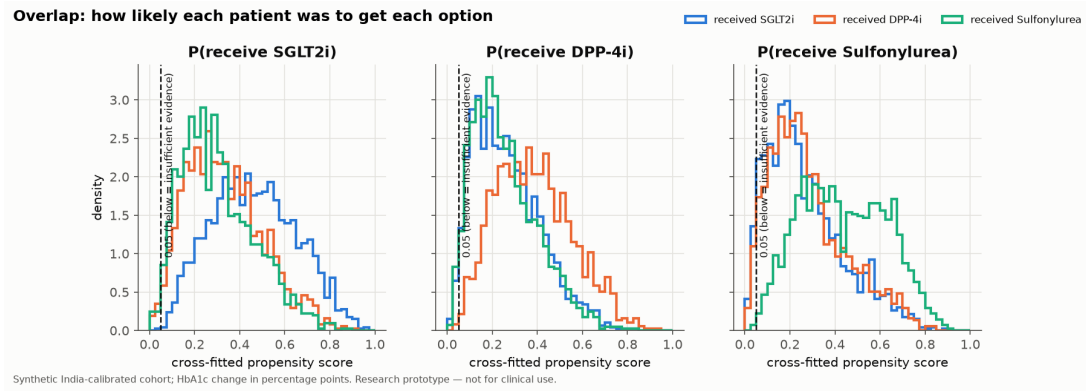


Figure 5.2: Propensity overlap for each drug; the dashed line marks the 0.05 abstention threshold

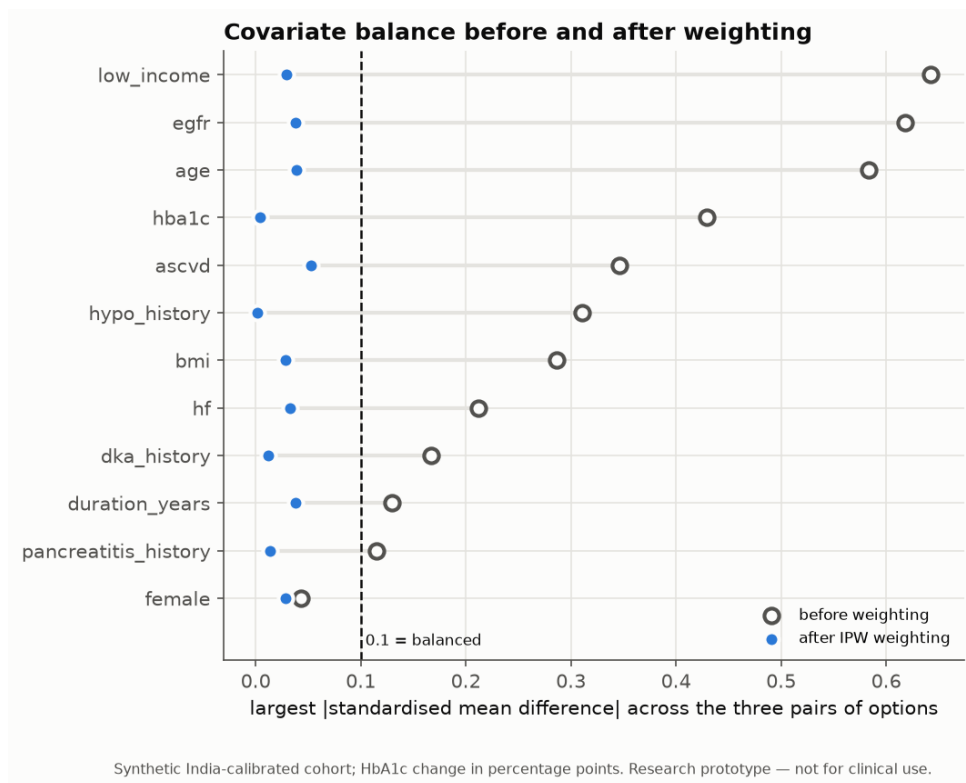


Figure 5.3: Covariate balance before and after weighting (target: standardised difference below 0.1 [1])

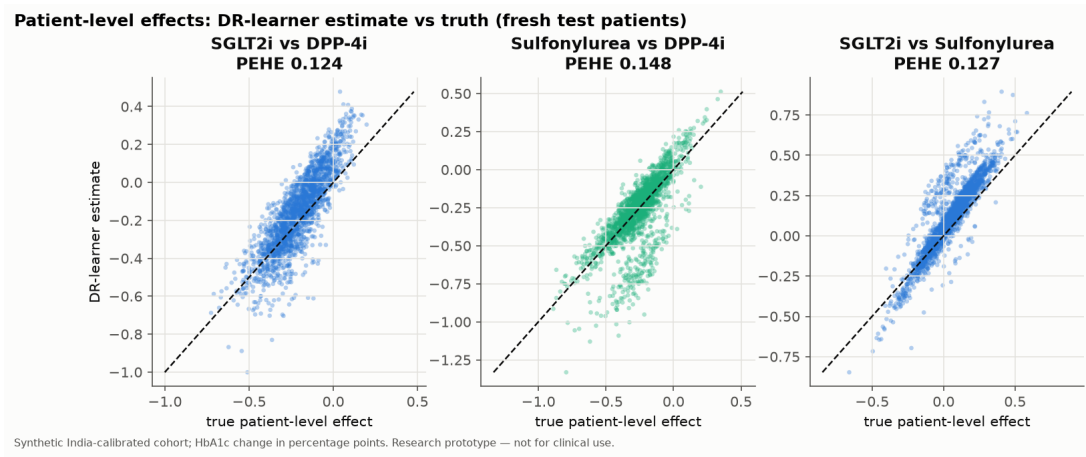


Figure 5.4: Patient-level effects estimated by the DR-learner against the true effects

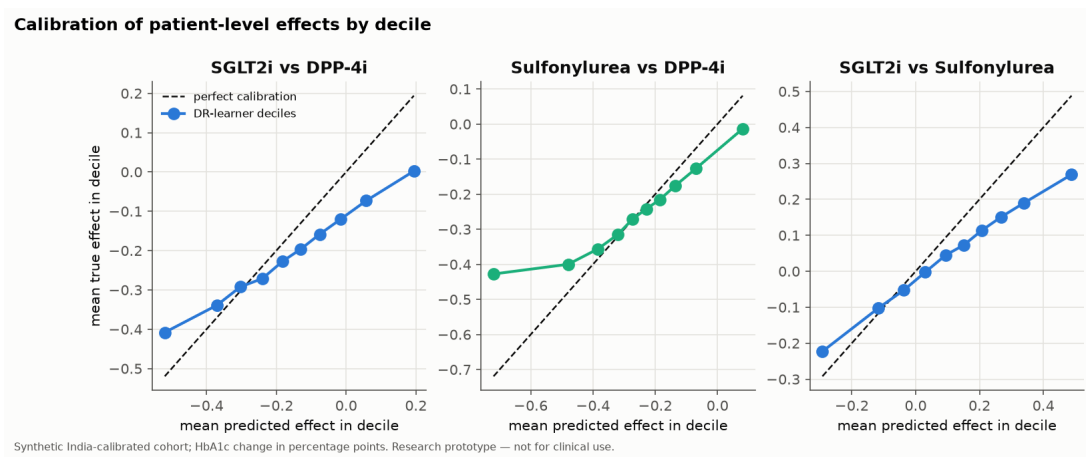


Figure 5.5: Calibration of patient-level effect estimates by decile

5.4 Secondary outcomes

The same cohorts, propensities, AIPW and DR-learner give two secondary outcomes for every estimated option: the 6-month weight change in kilograms and the risk of any hypoglycaemia, each with a 95% interval. They are generated from a separate random stream, so every HbA1c result above is unchanged. Over the 20 cohorts, AIPW estimated the weight difference between SGLT2i and sulfonyleurea as -3.33 kg against a truth of -3.33 kg, and the extra hypoglycaemia risk of sulfonyleurea over DPP-4i as 12.6 percentage points against a truth of 12.9; the DR-learner’s patient-level 95% intervals contained the truth for 94–96% (weight) and 93–94% (hypoglycaemia) of test patients. An option whose HbA1c interval is wider than 1.5 percentage points (team-set, to be confirmed by the clinician) is shown as “insufficient evidence (too uncertain)”.

5.5 RAG evaluation

The evidence layer searches only licence-cleared, team-confirmed documents: the WHO 2018 guideline on second- and third-line medicines (CC BY-NC-SA 3.0 IGO) and seven US FDA drug safety communications (78 passages). It combines BM25 and TF-IDF rankings

by reciprocal rank fusion, abstains when the best match is weak or the passages cover less than 60% of the question’s words, removes dose-like text from the index, and explains the answer by quoting the best-matching sentences with their passage numbers; answers written by a language model (Gemini free tier online, or a local Ollama model offline) are accepted only if every sentence passes a citation checker. Table 5.3 gives the results on 60 gold questions (45 answerable, 10 out of scope, 5 dose requests). The thresholds were tuned on the same questions, so the numbers are optimistic; abstention is below its target, and a medical embedding model, a reranker and a held-out question set are planned for October.

Table 5.3: RAG evaluation on the 60-question gold set (template explanations)

Measure	Value	Target
Recall@5	0.933	≥ 0.80
Answerable questions answered	0.978	—
Citation precision	1.000	≥ 0.95
Out-of-scope questions refused	0.800	≥ 0.95
Dose leaks	0	0

5.6 Screenshots of the System

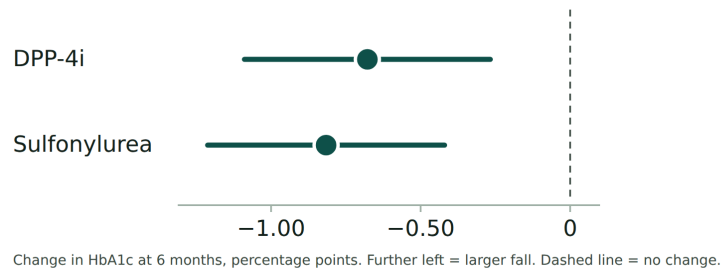


Figure 5.6: Demo: a typical patient with an estimate and 95% range for each option

BMI category: Obese (25 or above, Asian-Indian cut-off) · Request ID aa3a114ff56b · computed on this device

The three options (safety rules ran first)

SGLT2 inhibitor EXCLUDED e.g. dapagliflozin Not estimated: removed by a safety rule before the causal engine ran. R01 · Excluded (egfr < 45; rule verified): Not recommended to improve glycaemic control when eGFR < 45 mL/min/1.73 m2. KDIGO 2022 Rec 1.3.1 supports SGLT2i at eGFR ≥ 20 for kidney/heart protection - a different purpose; flag for the clinician. Source: FARXIGA (dapagliflozin) US prescribing information (DailyMed); KDIGO 2022 Diabetes in CKD — PI Sec 1 Limitations of Use, 2.2; KDIGO Rec 1.3.1 Cost: price unavailable	DPP-4 inhibitor ESTIMATE · CAUTION e.g. sitagliptin −0.68 (95% range −1.09 to −0.27) percentage points of HbA1c at 6 months · propensity 0.54 R04 · Caution (egfr < 45; rule verified): Renal dose adjustment required per label when eGFR < 45 mL/min/1.73 m2. This system never displays doses - consult the label. Source: JANUVIA (sitagliptin) US prescribing information (DailyMed) — Sec 2.2 R05 · Caution (pancreatitis_history = 1; rule verified): Not studied in patients with a history of pancreatitis; acute pancreatitis reported after marketing. Source: JANUVIA US prescribing information (DailyMed) — Sec 1; 5.1 Cost: price unavailable	Sulfonylurea ESTIMATE · CAUTION e.g. glimepiride −0.82 (95% range −1.21 to −0.42) percentage points of HbA1c at 6 months · propensity 0.23 R09 · Caution (egfr < 60; rule unverified): Kidney impairment increases hypoglycaemia risk; label advises a conservative start. eGFR cut-off 60 is team-set. Source: Glimepiride US prescribing information (DailyMed) — Sec 2.1; 8.6 Cost: price unavailable
--	--	--

Estimates with their 95% ranges**Fair comparisons**

Comparison	Difference	95% range
Sulfonylurea minus DPP-4i	−0.14	−0.71 to +0.43

Negative = the first option lowers HbA1c more. A range that crosses 0 means no clear difference.

Decision support only. The clinician decides.► **Assumptions behind these numbers**► **Structured Causal Output (JSON)**

Figure 5.7: Demo: an exclusion and cautions with their sources (eGFR 40, history of pancreatitis)

BMI category: Overweight (23 to below 25, Asian-Indian cut-off) · Request ID b7f4da37d5d8 · computed on this device

The three options (safety rules ran first)

SGLT2 inhibitor e.g. dapagliflozin	DPP-4 inhibitor e.g. sitagliptin	Sulfonylurea e.g. glimepiride
EXCLUDED	ESTIMATE · CAUTION	INSUFFICIENT EVIDENCE
Not estimated: removed by a safety rule before the causal engine ran. R01 · Excluded (egfr < 45; rule verified): Not recommended to improve glycaemic control when eGFR < 45 mL/min/1.73 m2. KDIGO 2022 Rec 1.3.1 supports SGLT2i at eGFR ≥ 20 for kidney/heart protection - a different purpose; flag for the clinician. Source: FARXIGA (dapagliflozin) US prescribing information (DailyMed); KDIGO 2022 Diabetes in CKD — PI Sec 1 Limitations of Use, 2.2; KDIGO Rec 1.3.1 Cost: price unavailable	−0.63 (95% range −0.85 to −0.41) percentage points of HbA1c at 6 months · propensity 0.81 R04 · Caution (egfr < 45; rule verified): Renal dose adjustment required per label when eGFR < 45 mL/min/1.73 m2. This system never displays doses - consult the label. Source: JANUVIA (sitagliptin) US prescribing information (DailyMed) — Sec 2.2 Cost: price unavailable	Too few similar patients received this option (propensity 0.006, below 0.05). A fair comparison is not possible. R07 · Caution (hypo_history = 1; rule verified): Sulfonylureas can cause severe hypoglycaemia; prior hypoglycaemia increases concern. Source: Glimepiride US prescribing information (DailyMed) — Sec 5.1 R08 · Caution (age ≥ 65; rule unverified): Older patients are at higher hypoglycaemia risk; label advises a conservative start. Age cut-off 65 is team-set. Source: Glimepiride US prescribing information (DailyMed) — Sec 2.1; 8.5 R09 · Caution (egfr < 60; rule unverified): Kidney impairment increases hypoglycaemia risk; label advises a conservative start. eGFR cut-off 60 is team-set. Source: Glimepiride US prescribing information (DailyMed) — Sec 2.1; 8.6 Cost: price unavailable

Estimates with their 95% ranges

DPP-4i

Change in HbA1c at 6 months, percentage points. Further left = larger fall. Dashed line = no change.

Decision support only. The clinician decides.

► **Assumptions behind these numbers**

► **Structured Causal Output (JSON)**

Figure 5.8: Demo: “insufficient evidence” shown instead of a guess (older patient, past hypoglycaemia)

Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

DiaCausal Synthetic data · The clinician decides **Local copy**

Try it **Evidence** Results Learn About

Evidence from licence-cleared sources

The **RAG** part of DiaCausal: it finds the passages of approved documents that best answer a question and shows each one with its citation. It never writes its own answer and never shows doses. Your question stays on this device.

Question

Can metformin be used with reduced kidney function (eGFR)?

Search the evidence

Metformin and kidney function Contrast imaging Lactic acidosis Not in the sources

5 passages, best match first

1. "Safety Announcement", page 1

Source S08: FDA Drug Safety Communications: metformin and reduced kidney function (8 Apr 2016); saxagliptin/alogliptin and heart failure (5 Apr 2016)

... The U.S. Food and Drug Administration (FDA) is requiring labeling changes regarding the recommendations for metformin-containing medicines for diabetes to expand metformin's use in certain patients with reduced kidney function. The current labeling strongly recommends against use of metformin in some patients whose kidneys do not work normally. We were ...

[Read the whole passage](#)

Figure 5.9: Evidence tab: cited passages from licence-cleared sources

SGLT2 inhibitor e.g. dapagliflozin	DPP-4 inhibitor e.g. sitagliptin	Sulfonylurea e.g. glimepiride
ESTIMATE	ESTIMATE	ESTIMATE
-0.93 (95% range -0.98 to -0.87)	-0.70 (95% range -0.76 to -0.64)	-0.94 (95% range -1.00 to -0.88)
percentage points of HbA1c at 6 months · propensity 0.43	percentage points of HbA1c at 6 months · propensity 0.25	percentage points of HbA1c at 6 months · propensity 0.32
Weight at 6 months -2.1 kg (95% range -2.3 to -2.0)	Weight at 6 months -0.4 kg (95% range -0.6 to -0.2)	Weight at 6 months +1.0 kg (95% range +0.8 to +1.2)
Any low sugar by 6 months 2.5% (95% range 0.8 to 4.2)	Any low sugar by 6 months 1.8% (95% range 0.1 to 3.4)	Any low sugar by 6 months 9.9% (95% range 5.7 to 14.1)
Cost: price unavailable	Cost: price unavailable	Cost: price unavailable
Evidence for this option	Evidence for this option	Evidence for this option

Figure 5.10: Secondary outcomes on each estimate: 6-month weight change and risk of any low blood sugar, each with a 95% range (typical patient)

Explanation (sentences quoted from the passages below)

- "We issued a Drug Safety Communication in May 2015 warning about the risk of ketoacidosis with SGLT2 inhibitors and alerting that we would continue to evaluate this safety issue." **[1]**
- "Health care professionals should assess for ketoacidosis and urinary tract infections in patients taking SGLT2 inhibitors who present with suggestive symptoms." **[1]**
- "This required enhanced pharmacovigilance study requests that manufacturers perform analyses of spontaneous postmarketing reports of ketoacidosis in patients treated with SGLT2 inhibitors, including specialized follow-up to collect additional information, for a period of 5 years." **[2]**
- "In addition to ketoacidosis and serious urinary tract infections, possible side effects of SGLT2 inhibitors include dehydration, kidney problems, increased cholesterol, and yeast infections." **[5]**

Numbers in brackets point to the passages below. The clinician decides.

Figure 5.11: RAG explanation: sentences quoted from the retrieved passages, each numbered to its source

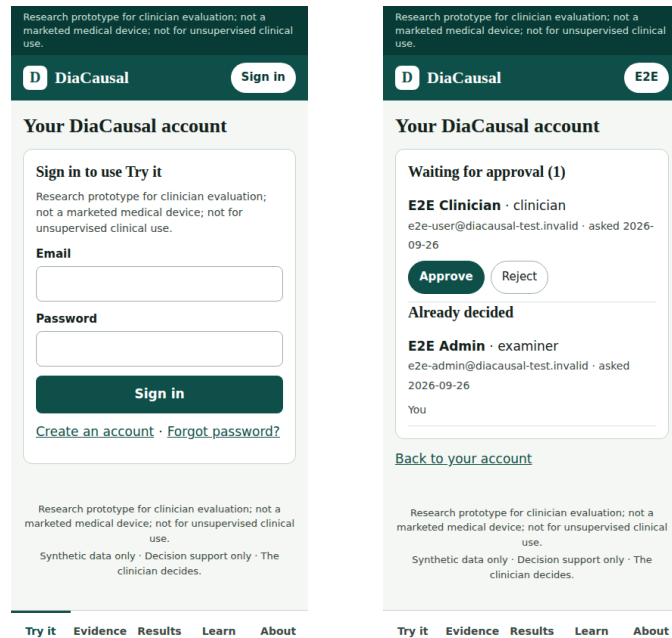


Figure 5.12: Phone website: sign-in (left) and the admin's approval screen with test accounts (right)

5.7 Sample Code (implementation)

The listings below are copied from the project repository (`diacausal_engine/`).

Listing 5.1: Safety rules engine (`guardrails.py`): rules are read from `rules.csv`, never hard-coded

```

1 class RulesError(ValueError):
2     """data/rules.csv has a row the engine cannot trust."""
3
4
5 @dataclass(frozen=True)
6 class Rule:
7     rule_id: str
8     arm: str
9     representative_molecule: str
10    field: str
11    op: str
12    value: float
13    action: str
14    message: str
15    source: str
16    section: str
17    status: str
18
19    def applies(self, patient: dict) -> bool:
20        if self.field not in patient or patient[self.field] is None:
21            # Fail closed: a rule we cannot check is a reason to stop, never to pass.
22            raise RulesError(f"{self.rule_id} needs '{self.field}', which was not given")
23        return OPS[self.op](float(patient[self.field]), self.value)
24
25    @property
26    def condition(self) -> str:
27        return f"{self.field} {OP_WORDS[self.op]} {self.value:g}"
28
29
30 @dataclass
31 class ArmVerdict:
32     """What the rules say about one option for one patient."""
33
34     arm: str
35     excluded: bool = False

```

```

36     fired: list[Rule] = field(default_factory=list)
37
38     @property
39     def cautions(self) -> list[Rule]:
40         return [r for r in self.fired if r.action == "CAUTION"]
41
42     @property
43     def exclusions(self) -> list[Rule]:
44         return [r for r in self.fired if r.action == "EXCLUDE"]
45
46     @property
47     def status(self) -> str:
48         if self.excluded:
49             return "excluded"
50         return "caution" if self.cautions else "no_rule_fired"
51
52
53 @dataclass(frozen=True)
54 class RuleTable:
55     rules: tuple[Rule, ...]
56     version: str
57
58     def apply(self, patient: dict) -> dict[str, ArmVerdict]:
59         verdicts = {arm: ArmVerdict(arm) for arm in ARMS}
60         for rule in self.rules:
61             if rule.applies(patient):
62                 v = verdicts[rule.arm]
63                 v.fired.append(rule)
64                 if rule.action == "EXCLUDE":
65                     v.excluded = True
66         return verdicts

```

Listing 5.2: AIPW scores (estimators.py)

```

1 def aipw_scores(Y: np.ndarray, T: np.ndarray, e: np.ndarray, mu: np.ndarray) -> np.ndarray:
2     """ $\phi_a = \mu_a(X) + 1\{T=a\} (Y - \mu_a(X)) / e_a(X)$ , for every patient and option, (n, 3).
3
4     The outcome model makes a guess; the second term corrects that guess using the
5     patients who really got option a, weighted by 1/e. If EITHER model is right, the
6     average of  $\phi$  is right -- hence "doubly robust".
7     """
8     got = (T[:, None] == np.arange(len(ARMS))[None, :]).astype(float)
9     return mu + got * (Y[:, None] - mu) / e
10
11
12 def aipw(phi: np.ndarray, z: float) -> list[Estimate]:
13     return _from_influence("AIPW", phi, phi.mean(axis=0), z)

```

Listing 5.3: DR-learner with HC3 robust intervals (estimators.py)

```

1 class DRLearner:
2     """Kennedy's DR-learner with a linear second stage and HC3 robust standard errors.
3
4     Plain English:  $\phi$  is a noisy but fair "what would have happened under option a" for
5     every patient. Regressing  $\phi$  on the patient's details averages out the noise and
6     gives a formula; plugging THIS patient's details into the formula gives their
7     personalised estimate. The robust standard error of that prediction gives the 95%
8     interval:  $x' \beta +/ - 1.96 * \sqrt{x' V x}$ .
9     """
10
11     def fit(self, X: np.ndarray, phi: np.ndarray) -> DRLearner:
12         self.mean_ = X.mean(axis=0)
13         self.scale_ = np.where(X.std(axis=0) > 0, X.std(axis=0), 1.0)
14         B = self._basis(X)
15         pseudo = pseudo_outcomes(phi)
16         bread = np.linalg.pinv(B.T @ B)
17         leverage = np.einsum("ij,jk,ik->i", B, bread, B)
18         self.beta_ = bread @ B.T @ pseudo # (p, 6)
19         resid = pseudo - B @ self.beta_
20         adj = resid / (1.0 - np.clip(leverage, 0, 0.99))[:, None] # HC3
21         self.cov_ = np.stack([bread @ (B.T * adj[:, k] ** 2) @ B @ bread for k in range(len(TARGETS))])
22         return self
23

```

```

24 def _basis(self, X: np.ndarray) -> np.ndarray:
25     Xs = (np.atleast_2d(X) - self.mean_) / self.scale_
26     return np.column_stack([np.ones(len(Xs)), Xs])
27
28 def predict(self, X: np.ndarray, z: float) -> dict[str, np.ndarray]:
29     """{target: (n, 3) array of [estimate, low, high]}."""
30     B = self._basis(X)
31     out = {}
32     for k, target in enumerate(TARGETS):
33         value = B @ self.beta[:, k]
34         se = np.sqrt(np.einsum("ij,jk,ik->i", B, self.cov_[k], B))
35         out[target] = np.column_stack([value, value - z * se, value + z * se])
36     return out

```

5.8 Evaluation matrix

Table 5.4 lists how the complete system will be judged. Targets marked L come from the literature; targets marked T were set by the team. The last column gives the mid-semester result where one exists (synthetic data).

Table 5.4: Evaluation matrix

Part	Metric	Meaning	Target	Mid-semester result
Causal	PEHE per drug pair	Error of patient-level effects	Lower than simple learners (T)	DR 0.105–0.117 vs T-learner 0.184–0.192; naive 0.169–0.214
Causal	Bias and RMSE of average effects	Over repeated cohorts	$ \text{bias} \leq 0.1$ at $n = 5,000$ (T)	AIPW $ \text{bias} \leq 0.005$; RMSE 0.023–0.030
Causal	95% interval coverage	Share of intervals containing the truth	90–97% (T)	AIPW 90–95%; DR-learner 92.7–94.7%
Causal	Policy regret	Loss from choosing the estimated best option	Lower than naive (T)	0.011 vs 0.062
Causal	Covariate balance	Largest SMD after weighting	Below 0.1 (L [1])	0.70 $\rightarrow$ 0.08
Causal	Overlap and abstention	Share of answers refused	Reported (T)	2.5% of patient–option pairs
Causal	Refutation tests	Placebo, random cause, subsets	Placebo near 0; stable (L [22])	9/9 passed
Causal	E-value	Strength of hidden confounding needed	Reported (L)	1.57–2.15

Table 5.4: Evaluation matrix

Part	Metric	Meaning	Target	Mid-semester result
Safety rules	Rule tests	Rules unit-tested and passing	100% (T)	100% (201 tests pass)
Safety rules	Unsafe options shown	On 50 or more test cases	0 (T)	0: an excluded option is never estimated (checked on 155 test patients); formal set in October
RAG	Recall@5	Right passage in the top 5	At least 0.80 (T)	0.933 (60 gold questions)
RAG	Citation precision	Citations that support the sentence	At least 0.95 (T)	1.000 (quoted explanations)
RAG	Faithfulness	Claims supported by passages	At least 0.90 (T)	October, for the Gemini and Ollama modes (RAGAS [25])
RAG	Abstention accuracy	Out-of-scope questions refused	At least 0.95 (T)	0.800 (not yet met)
RAG	Free-text doses	Doses written by the system	0 (T)	0 on 60 questions (dose check on every reply)
Clinician	Agreement	Cohen’s κ on 60 cases	At least 0.61 (L [32])	October (vignette study)
Clinician	Usability	System Usability Scale	At least 68 (L [31])	October (survey)
System	Latency	95th-percentile response time	Under 3 s (T)	Website answers on the device instantly; load test in October
System	Audit completeness	Requests logged	100% (T)	Every engine request logged (tested)
Licences	Licence records	Sources with a licence category	100% (T)	100% (21 sources in the registry)

5.9 Discussion: work completed and work planned

By the mid-semester review every core part of DiaCausal works on its own and is tested on synthetic data: the secure chat application; the causal engine with its benchmark, refutation checks and secondary outcomes; a live phone website with approved accounts; and a RAG layer that searches licence-cleared WHO and FDA text and explains answers with checked citations. Since 29 September the chat application runs the causal engine and the evidence search inside its six-stage pipeline. The guide signed the required 25%

on 23 September; the Project Interaction Sheet’s 40% and 60% checks were met on 28 and 29 September, and the team’s weighted estimate for the whole project is about 75%. Because the true effects are synthetic, good recovery shows that the estimators are correct, not that the system is valid for real patients; that is why a clinician review and a real-data check are part of the plan. The remaining work is a medical embedding model and reranker with a held-out question set (target 3 October), a real-data adapter and evaluation tools (target 6 October), the full evaluation in Table 5.4, the clinician review and usability survey, and the research paper.

Chapter 6

Conclusion & Future Scope

6.1 Conclusion

DiaCausal treats the choice of a second-line diabetes drug as a question about what would happen to one patient under each option, and answers it with causal inference placed between deterministic safety rules and cited explanation. By the mid-semester review the team has fixed the scope and data strategy, specified the requirements and design, and implemented and tested the causal engine on an India-calibrated synthetic cohort with known true effects: the corrected estimators remove the bias of the naive comparison, their intervals contain the truth close to 95% of the time, and the system refuses to guess when similar patients are too few. The engine is available on any phone through an installable website with approved accounts, and the first retrieval layer already returns cited passages from licence-cleared sources. Every estimate has an interval, and the clinician always makes the decision.

6.2 Future Scope

- **Plan to 100% (Project Interaction Sheet):** the sheet checks 40% implementation on 7 October, 60% on 14 October and 80% on 21 October. The 40% and 60% checks were met early, on 28 and 29 September; section (3), the causal engine and RAG inside the chat interface, is done. The team aims to finish section (4), a dense medical embedding model with a reranker (target 3 October), a real-data adapter and the evaluation tooling, by 6 October. Guide evaluation follows on 26–31 October, the final synopsis presentation in November, and in December the 100% sign-off with the research paper (format, plagiarism check, language and flow). In January and February 2027 come the complete project evaluation, PCUBE (poster, presentation and model), the paper revision and submission, and the black book. A clinician review of the gold questions and realistic cases with a usability survey runs through October. (Secondary outcomes, the consultation summary and the first RAG explanations and evaluation were completed ahead of plan in September.)
- **Validation on real data:** de-identified Indian clinic records and cohorts such as ICMR-INDIAB, only with ethics committee approval, followed by recalibration.
- **More treatments:** GLP-1 receptor agonists and insulin.
- **Offline hospital installation:** the same system installed on a hospital’s own computers, with a local language model and no public access.
- **Prospective evaluation and reporting:** following DECIDE-AI [33], TRI-POD+AI [34] and TARGET [15], in line with the ICMR ethical guidelines for AI in healthcare [35].

Intended use: Research prototype for clinician evaluation; not a marketed medical device; not for unsupervised clinical use.

References

- [1] P. C. Austin, “Balance diagnostics for comparing the distribution of baseline covariates between treatment groups in propensity-score matched samples,” *Statistics in Medicine*, vol. 28, no. 25, pp. 3083–3107, 2009.
- [2] R. M. Anjana, R. Unnikrishnan, M. Deepa, *et al.*, “Metabolic non-communicable disease health report of India: the ICMR-INDIAB national cross-sectional study (ICMR-INDIAB-17),” *The Lancet Diabetes & Endocrinology*, vol. 11, no. 7, pp. 474–489, 2023.
- [3] International Diabetes Federation, *IDF Diabetes Atlas*. Brussels: IDF, 11th ed., 2025.
- [4] B. M. Shields, J. M. Dennis, C. D. Angwin, *et al.*, “Patient stratification for determining optimal second-line and third-line therapy for type 2 diabetes: the TriMaster study,” *Nature Medicine*, vol. 29, pp. 376–383, 2023.
- [5] A. Tsapas, I. Avgerinos, T. Karagiannis, *et al.*, “Comparative effectiveness of glucose-lowering drugs for type 2 diabetes: a systematic review and network meta-analysis,” *Annals of Internal Medicine*, vol. 173, no. 4, pp. 278–286, 2020.
- [6] S. C. Palmer, D. Mavridis, A. Nicolucci, *et al.*, “Comparison of clinical outcomes and adverse events associated with glucose-lowering drugs in patients with type 2 diabetes: a meta-analysis,” *JAMA*, vol. 316, pp. 313–324, 2016.
- [7] R. T. Sutton, D. Pincock, D. C. Baumgart, *et al.*, “An overview of clinical decision support systems: benefits, risks, and strategies for success,” *npj Digital Medicine*, vol. 3, p. 17, 2020.
- [8] P. Lewis, E. Perez, A. Piktus, *et al.*, “Retrieval-augmented generation for knowledge-intensive NLP tasks,” in *Advances in Neural Information Processing Systems 33*, pp. 9459–9474, 2020.
- [9] C. Zakka, R. Shad, A. Chaurasia, *et al.*, “Almanac – retrieval-augmented language models for clinical medicine,” *NEJM AI*, vol. 1, no. 2, 2024.
- [10] J. M. Dennis, K. G. Young, A. P. McGovern, *et al.*, “Development of a treatment selection algorithm for SGLT2 and DPP-4 inhibitor therapies in people with type 2 diabetes: a retrospective cohort study,” *The Lancet Digital Health*, vol. 4, no. 12, pp. e873–e883, 2022.
- [11] L. M. Güdemann, K. G. Young, P. Cardoso, *et al.*, “Validation of an algorithm for selection of SGLT2 and DPP4 inhibitor therapies in people with type 2 diabetes across major UK ethnicity groups: a retrospective cohort study,” *The Lancet Regional Health – Europe*, vol. 61, p. 101547, 2026.

- [12] D. B. Rubin, “Estimating causal effects of treatments in randomized and nonrandomized studies,” *Journal of Educational Psychology*, vol. 66, no. 5, pp. 688–701, 1974.
- [13] J. Pearl, *Causality: Models, Reasoning, and Inference*. Cambridge University Press, 2nd ed., 2009.
- [14] M. A. Hernán and J. M. Robins, *Causal Inference: What If*. Chapman & Hall/CRC, 2020.
- [15] A. G. Cashin, H. J. Hansford, M. A. Hernán, *et al.*, “Transparent reporting of observational studies emulating a target trial – the TARGET statement,” *JAMA*, vol. 334, no. 12, pp. 1084–1093, 2025.
- [16] S. R. Künzel, J. S. Sekhon, P. J. Bickel, and B. Yu, “Metalearners for estimating heterogeneous treatment effects using machine learning,” *PNAS*, vol. 116, no. 10, pp. 4156–4165, 2019.
- [17] S. Wager and S. Athey, “Estimation and inference of heterogeneous treatment effects using random forests,” *Journal of the American Statistical Association*, vol. 113, no. 523, pp. 1228–1242, 2018.
- [18] E. H. Kennedy, “Towards optimal doubly robust estimation of heterogeneous causal effects,” *Electronic Journal of Statistics*, vol. 17, no. 2, pp. 3008–3049, 2023.
- [19] V. Chernozhukov, D. Chetverikov, M. Demirer, E. Duflo, C. Hansen, W. Newey, and J. Robins, “Double/debiased machine learning for treatment and structural parameters,” *The Econometrics Journal*, vol. 21, no. 1, pp. C1–C68, 2018.
- [20] J. L. Hill, “Bayesian nonparametric modeling for causal inference,” *Journal of Computational and Graphical Statistics*, vol. 20, no. 1, pp. 217–240, 2011.
- [21] U. Shalit, F. D. Johansson, and D. Sontag, “Estimating individual treatment effect: generalization bounds and algorithms,” in *Proc. 34th ICML, PMLR 70*, pp. 3076–3085, 2017.
- [22] A. Sharma and E. Kiciman, “DoWhy: An end-to-end library for causal inference.” arXiv:2011.04216, 2020.
- [23] Microsoft Research, “EconML: A Python package for ML-based heterogeneous treatment effects estimation.” <https://github.com/py-why/EconML>, 2019.
- [24] G. Xiong, Q. Jin, Z. Lu, and A. Zhang, “Benchmarking retrieval-augmented generation for medicine,” in *Findings of the Association for Computational Linguistics: ACL 2024*, pp. 6233–6251, 2024.
- [25] S. Es, J. James, L. Espinosa-Anke, and S. Schockaert, “RAGAS: Automated evaluation of retrieval augmented generation,” in *Proc. EACL 2024: System Demonstrations*, pp. 150–158, 2024.
- [26] P. Quaranta and R. Pietrantuono, “Exploring causal modeling to enhance diabetes prediction and management.” IEEE Xplore, document 11058734, 2025.

-
- [27] U. Hasan, M. O. Gani, M. Adibuzzaman, and S. Kethireddy, “Leveraging causal AI for robust treatment policy formulation: an ensemble causal tree approach with expert insights in critical care decision-making.” IEEE Xplore, document 11038719, 2025.
 - [28] X. Song, X. Zan, Z. Fu, and X. Ding, “Causal inference based hierarchical regression model for cuffless blood pressure estimation,” in *Proc. 46th Annual Int. Conf. IEEE EMBC*, pp. 1–4, 2024.
 - [29] A. Misra, P. Chowbey, B. M. Makkar, *et al.*, “Consensus statement for diagnosis of obesity, abdominal obesity and the metabolic syndrome for Asian Indians and recommendations for physical activity, medical and surgical management,” *Journal of the Association of Physicians of India*, vol. 57, pp. 163–170, 2009.
 - [30] International Institute for Population Sciences (IIPS) and NPHCE, MoHFW and Harvard T. H. Chan School of Public Health and University of Southern California, “Longitudinal ageing study in india (LASI) wave 1, 2017–18, india report,” tech. rep., International Institute for Population Sciences, Mumbai, 2020.
 - [31] J. Brooke, “SUS: A quick and dirty usability scale,” in *Usability Evaluation in Industry*, pp. 189–194, Taylor & Francis, 1996.
 - [32] J. R. Landis and G. G. Koch, “The measurement of observer agreement for categorical data,” *Biometrics*, vol. 33, no. 1, pp. 159–174, 1977.
 - [33] B. Vasey, M. Nagendran, B. Campbell, *et al.*, “Reporting guideline for the early-stage clinical evaluation of decision support systems driven by artificial intelligence: DECIDE-AI,” *Nature Medicine*, vol. 28, pp. 924–933, 2022.
 - [34] G. S. Collins, K. G. M. Moons, P. Dhiman, *et al.*, “TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods,” *BMJ*, vol. 385, p. e078378, 2024.
 - [35] Indian Council of Medical Research, “Ethical guidelines for application of artificial intelligence in biomedical research and healthcare,” 2023.

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Appendix A : Timeline Chart

Both charts follow the department's Project Interaction Sheet (Second Half of 2026 and First Half of 2027). Figure A.1 lists every entry of the sheet with its date and status; weeks 0–7 are signed by the guide. Figure A.2 shows the same plan as a Gantt chart. As the sheet requires, implementation is split into four sections: (1) the causal engine core and the chat application base, (2) the evidence search (RAG), secondary outcomes and the website, (3) the causal engine and RAG inside the chat application, and (4) better retrieval, a real-data adapter and the evaluation tools. The charts contain only the sheet's own rows and wording. The sheet's implementation checks are 25% by 23 September (signed), 40% by 7 October (met on 28 September), 60% by 14 October (met on 29 September) and 80% by 21 October (team target 3 October); the team's target for all four sections is 6 October. Diamonds mark Synopsis Presentations I (12 August) and II (30 September 2026); the dashed line marks today.

Timeline chart: Project Interaction Sheet, Group 28 (2026–27)



Figure A.1: Timeline chart: every entry of the Project Interaction Sheet (2026–27)

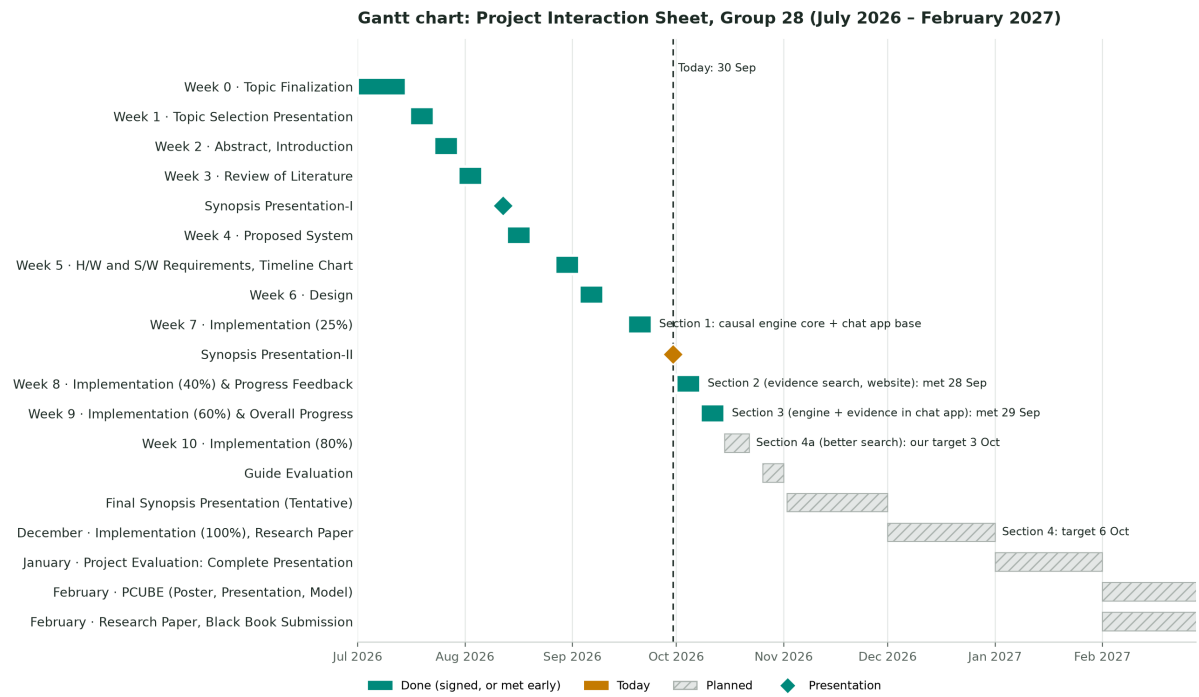


Figure A.2: Gantt chart from the Project Interaction Sheet (July 2026–February 2027)

Appendix B : Publication Details

Status: in preparation. The paper will be written after the results are frozen on 16 October 2026, in IEEE conference format.

Candidate venues (to be confirmed with the guide):

- IEEE EMBC 2027, the 49th Annual International Conference of the IEEE Engineering in Medicine and Biology Society, Singapore, 11–15 July 2027 (paper deadline not yet announced).
- IEEE ICHI 2027, the IEEE International Conference on Healthcare Informatics (regular deadline not yet announced).

Reporting: the TARGET checklist for the emulated-trial framing [15] and TRIPOD+AI items where applicable [34]. Use of AI tools in preparing the paper will be disclosed in its acknowledgment, as IEEE requires.